

MarketMind AI

Software Product Requirements Documentation — complete specification set (00–30)

AI Strategic Intelligence Platform | Version 1.1 — reviewer feedback incorporated | 04 July 2026

New in 1.1: Decision Workspace (30) · Market Timeline · visible reasoning chains (Signal → Impact → Reason → Recommendation)
· evidence-derived conviction scores on recommendations · Daily CEO Brief · command-center UI · streamed-reasoning onboarding

MVP constraints: free tiers · open/public data first · robots.txt & ToS enforced in code · paid APIs pluggable without refactoring · finish V1, polish first

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Status	Approved
Owner	Product
Last updated	2026-07-04
Inherited by	Every document in this set (01-29)

MarketMind AI is not a dashboard. It is not a tracker. It is not an analytics platform.

MarketMind AI is an AI Strategic Intelligence Platform. Its purpose is not to show founders information. Its purpose is to help founders make better strategic decisions.

The one test

Every feature, screen, schema field, and prompt must pass a single test:

Does this help the founder decide what to build, what to ignore, and what to do next?

If it cannot answer that question, it does not ship. It is simplified or removed.

The problem we exist to remove

Founders drown in tabs — GitHub, Product Hunt, Reddit, X, blogs, docs, pricing pages, release notes, AI announcements, hiring pages. By the time they finish reading, the information is stale. They do not need more information. They need someone who reads everything and answers:

- 1. **What changed?**
- 2. **Why does it matter?**
- 3. **Should I care?**
- 4. **What should I do?**

MarketMind AI is that analyst — standing, always-on, and honest about what it knows versus what it infers. And it does not stop at answers: every recommendation ends in a decision the founder can take in one click — save it, roadmap it, or ignore it — so the product changes what people *do*, not just what they read.

Non-negotiable beliefs

- 1. **Recommendation over raw data.** Raw feeds are a fallback view, never the default. Every surfaced signal answers "so what" before "what."
- 2. **Trust through epistemic honesty.** Every insight is labeled: **Verified Public Information, AI Inference, or Reasoned Recommendation.** Speculation is never presented as fact.
- 3. **Zero-effort by default.** One input — the user's website URL. Everything else is discovered, not configured. Configuration exists, layered on good defaults.
- 4. **Signal-to-noise is the product.** Success is showing the 5 things that mattered, not the 500 things we collected.
- 5. **Respect the web.** Public, non-authenticated sources only. robots.txt and terms of service are enforced in code, not policy.
- 6. **Free-core, paid-enhanced.** The complete loop works on free/public data and free-tier infrastructure. Paid providers plug in behind interfaces without refactoring.

What we refuse to be

- A generic scraper or data broker.
- A real-time trading/sentiment tool. Our timescale is days and weeks.
- A CRM.
- A substitute for legal or IP diligence.
- An autonomous decision-maker. Every recommendation is advisory. The product never acts on the user's behalf.

Long-term direction

MarketMind AI becomes to strategic decisions what Copilot is to code: the first tab a founder opens each morning — not because it tells them everything, but because it tells them **exactly what matters**.

01 — Product Vision

Status	Approved (Rev. 2 — amended per Vision Review and MVP constraints)
Owner	Product, Architecture, AI Engineering
Depends on	00_PRODUCT_MANIFESTO

1. Mission

MarketMind AI understands a company from nothing more than its website, automatically discovers its real competitors, continuously monitors the public signals that matter, interprets what those signals mean for *this* company, and converts that interpretation into specific, scheduled, decision-ready recommendations.

Four verbs, each mapping to a subsystem:

Verb	Subsystem	Document
Understand	Company Understanding Engine	11
Discover	Competitor Discovery Engine	11
Monitor	Signal Collection Engine	12
Interpret → Recommend	Analysis + Report Engines	13, 14
Decide	Decision Workspace	30

A proposed feature that serves none of these verbs is scope creep → route to 25_ROADMAP (future) instead of building.

2. MVP constraints (binding on all documents)

- Runs on **free tiers** wherever possible.
- Open/public data sources first**; paid data is an optional enhancement, never a dependency.
- robots.txt and ToS respected**, enforced in the fetch layer (see 12, 21).
- Architecture allows **paid APIs to be added later without major refactoring** (provider interfaces — see 08, 10, 28).
- Built for **learning, demonstration, and portfolio quality first**; scale optimization is deferred until user interest is validated.

3. The three-year picture

The durable asset is **not** the breadth of collected data (commoditizes) and **not** the reasoning layer alone (rented from model providers). The durable assets are:

- Proprietary longitudinal signal history** per competitor — compounds with time, cannot be backfilled by late entrants (bootstrapped at day one via Wayback Machine backfill — see 12).
- The feedback flywheel** — user judgments ("this mattered / this didn't") tune significance thresholds per segment.
- Purpose-built compliant collectors** for the signal categories that survive ToS reality.

4. Target users (detail in 03)

- Solo / early-stage founder** — primary. The wedge persona: zero-friction, zero-upkeep, price-sensitive.
- Product manager at a growth-stage company** — primary for exports/reports; the future monetization anchor.
- Investor / analyst** — secondary; bursty, on-demand analysis flows.

Explicit non-targets: enterprise CI teams with dedicated analysts; agencies running CI for many clients.

5. Core value proposition

"Tell us your website. We'll tell you who's coming for you, what they're doing, and what to do about it — on your schedule, without you lifting a finger."

Four moments that must each feel effortless (quality bars in 11, 12, 13, 14):

- Onboarding** — URL in → company understood → credible ranked competitor list with reasoning, in minutes.
- Baseline** — an immediate snapshot intelligence report (not a change report) so day one has value.
- Monitoring** — signals accumulate silently and correctly.
- Insight & Action** — a small number of things that matter, each with a visible reasoning chain (Signal → Impact → Reason → Recommendation) and evidence label.
- Decision** — every recommendation carries one-click actions (Save as Opportunity · Ignore · Create Feature · Add to Roadmap · Review Later) so the product changes what the founder *does*, not just what they read (30).

6. Verified vs. inferred: the product-wide contract

Three labels, enforced in schema (09), API (20), UI (17), and prompts (10):

- Verified Public Information** — directly observed from a primary source, with URL + timestamp.
- AI Inference** — a conclusion drawn by combining observations.
- Reasoned Recommendation** — advisory guidance derived from inferences.

The UI must never render an inference with the same visual weight as a verified fact.

7. Success metrics

MVP North Star: end-to-end demo quality on 5–10 real companies — a complete, honest, compliant loop — plus legibility of the architecture to a technical reviewer.

Post-validation North Star: Weekly Actioned Insights per Active Account.

Supporting: discovery acceptance rate; time-to-first-insight; report engagement; user-reported false-positive rate; churn correlated with monitoring gaps (requires collection-health telemetry from day one — see 22).

8. Known deferrals (deliberate, per MVP constraints)

Moat commercialization, persona monetization, and per-account COGS modeling are deferred until validation. See 29_MVP_LIMITATIONS for the complete honest list.

02 — Product Requirements Document

Status	Approved
Owner	Product
Depends on	00, 01

1. Scope model

Phase	Definition
V1 (MVP)	Free-tier, public-data, portfolio-grade. Complete loop for the three A-grade signal categories.
V2	Paid-provider slots activated (funding, firmographics, reviews), team features, more channels.
V3	Feedback-trained significance models, multi-language, integrations (Slack/CRM), API for customers.

2. V1 feature set

2.1 Onboarding & understanding

- Sign in (email magic link or GitHub OAuth). No multi-step onboarding wizard.
- Single required input: company website URL.
- **Company Understanding Engine** extracts: industry, business model, target audience, products, features, pricing, technology, keywords, value proposition, customer segment, company size, market positioning. All fields user-editable afterwards.

2.2 Competitor discovery

- Automatic discovery from public information + AI reasoning. No manual searching required.
- Each candidate carries a **confidence score** and visible reasoning ("why we think this is a competitor").
- User can accept, reject, or manually add competitors. Target: ≥70% acceptance rate on suggestions for well-documented SaaS.

2.3 Baseline intelligence report (day one)

Snapshot, not change report. Sections: market overview, competitor overview, technology comparison, pricing comparison, feature comparison, AI adoption, engineering activity, hiring trends, customer sentiment (where compliant sources exist), SWOT, threat level, opportunity areas, missing features, strategic recommendations. Historical depth via Wayback backfill (12).

2.4 Continuous monitoring (V1 signal categories)

Category	V1 status
Website / product / pricing changes	Full
Hiring signals (public ATS boards + careers pages)	Full
Blog / content / release notes	Full
Engineering activity (GitHub public)	Full
Funding / business news (free news + EDGAR)	Partial, labeled
Technology ecosystem intelligence (model-provider announcements)	Full (blogs/changelogs/status feeds)
Customer sentiment	Narrow slice (HN Algolia; Reddit within free-tier terms). G2/Capterra excluded (ToS).
Verified firmographics	Excluded — provider stub visible in UI as "requires data partner"

2.5 Intelligence layer

Every signal passes: Raw Event → Context → Business Impact → Strategic Meaning → Recommendation (13). Raw data is available as a fallback view only.

2.6 Reports

Daily Digest, Weekly Strategic Report, Monthly Market Report, Quarterly Competitive Review (14). Every report answers: what happened, what changed, why it matters, what should I do.

2.7 Search

Conversational search over collected evidence: "Analyze Notion", "Compare us with Linear", "What pricing changes happened this month?" (16).

2.8 Notifications

Email + Telegram in V1; HH:MM-precision scheduling with timezone/DST handling; per-category preferences; priority filtering; smart digest grouping; quiet hours / weekend pause / vacation mode / snooze; critical-override option (15).

2.9 Decision Workspace (30)

One-click decision actions on every recommendation; Inbox / Opportunities / Roadmap-lite (Considering → Planned → Building) / append-only Decision log. GitHub Issues export is a visible V2 slot.

2.10 Market Timeline

"Show me everything that happened during the last 6 months" — a chronological, filterable view over all signals and insights (per competitor, per category, or whole market), rendered as a month-grouped timeline. Powered directly by the existing `signals/insights` time series plus Wayback backfill, so it is deep on day one.

2.11 Daily CEO Brief

The Daily Digest is upgraded to a CEO Brief format (14): personalized greeting, yesterday's confirmed events as a checklist, one prioritized Today's Recommendation with market-urgency level and an evidence-derived conviction indicator listing its supporting signal families.

2.12 Trust surface

Three-label evidence system rendered distinctly everywhere (17, 18).

3. Out of scope for V1

See 29_MVP_LIMITATIONS — the authoritative list. Highlights: no paid data APIs, no autonomous actions, no WhatsApp/Slack/Discord/push, no multi-language, no team seats, no numeric confidence percentages in UI (evidence buckets instead).

4. Release criteria (V1 "done")

- 1. URL → confirmed competitor set in < 10 minutes for a known SaaS.
- 2. Baseline report generated within 30 minutes of onboarding, containing ≥ 3 genuinely longitudinal facts (Wayback-derived).
- 3. Seven consecutive days of scheduled collection with zero manual intervention and a visible collection-health log.
- 4. A weekly digest sent to email and Telegram at the user's configured HH:MM local time.
- 5. Every insight in every surface carries one of the three evidence labels.
- 6. Demo script: 5-minute end-to-end run on a company the audience knows.
- 7. Written coverage post-mortem across 5–10 tracked companies.
- 8. Decision Workspace round-trip: recommendation → Save as Opportunity → appears on Roadmap-lite → visible in Decision log with full evidence chain.
- 9. Market Timeline renders ≥ 6 months of history (backfill-powered) for a demo competitor set.

03 — User Personas

Status	Approved
Owner	Product
Depends on	01

P1 — The Solo Founder (Primary, wedge)

Ravi, 24, technical founder of a 3-person dev-tools startup. - Runs competitive analysis in the gaps between everything else. 30+ tabs daily. - **Needs:** to know what changed, whether it matters, and what to do — in under 5 minutes a day. - **Will not tolerate:** manual upkeep, configuration ceremony, raw-data firehoses, confident nonsense. - **Churn trigger:** one wrong "fact" presented without a source; any week the tool requires maintenance. - **Design consequences:** one-URL onboarding; opinionated defaults; evidence labels; digest over dashboard.

P2 — The Growth-Stage PM (Primary for reports; future monetization anchor)

Meera, 31, PM at a Series-B SaaS. - Must justify roadmap and pricing decisions to leadership with competitor evidence. - **Needs:** exportable, presentable, sourced reports; comparison tables; a defensible evidence trail. - **Design consequences:** report export, citation-per-claim, weekly strategic report shaped for forwarding.

P3 — The Investor / Analyst (Secondary)

Arjun, 35, seed-stage investor. - Bursty usage: evaluating one company or sector at a time, then dormant. - **Needs:** on-demand baseline reports and conversational analysis ("Analyze X"), not continuous monitoring. - **Design consequences:** ad-hoc analysis flows must work without long-lived monitoring setup (16).

Anti-personas (explicitly not designed for)

- Enterprise CI analysts needing custom taxonomies, battlecards, and workflow admin.
- Agencies operating CI for many client tenants.
- Anyone requiring private/authenticated data collection — the product will never do this.

04 — User Journeys

Status	Approved
Owner	Product, Design
Depends on	02, 03

J1 — Onboarding (target: < 10 minutes to confirmed competitor set)

1. Sign in (magic link / GitHub). No wizard.
2. Enter company URL. Single field, single action.
3. **Understanding step (automated, ~1-2 min):** the system fetches and characterizes the user's own site through the standard collection pipeline. A structured company profile is shown; every field editable.
4. **Discovery step:** ranked competitor candidates appear with confidence + one-line reasoning each. User accepts / rejects / adds. Low-confidence candidates are framed honestly: "2 confident matches, 5 guesses — help us calibrate."
5. **Baseline report** begins generating immediately after confirmation; user is told when it will be ready and on which channel.

J2 — The daily 5 minutes

1. 08:00 (user-chosen HH:MM, local tz): Daily Digest arrives on chosen channel(s).
2. Digest leads with the AI Strategic Summary (3-5 sentences), then grouped events by category, each with evidence label and "so what."
3. One tap/click opens the insight detail: sources, timeline, related signals, recommendation.
4. User marks 🗑️ useful / 🗑️ didn't matter → feeds the significance thresholds (13).

J3 — The weekly decision

Monday (or user-chosen day): Weekly Strategic Report. Structured for forwarding to a co-founder/leadership: what happened → what changed → why it matters → recommended actions, each action traceable to labeled evidence.

J4 — Ad-hoc analysis (P3 flow)

User types "Analyze Notion" or "Compare us with Linear" into Search. System returns a structured answer synthesizing stored evidence, clearly separating verified facts from inference (16). If the entity isn't tracked, a one-off analysis runs within quota.

J5 — Configuring notifications

Settings → Notifications: channels (email/Telegram), HH:MM time picker, frequency, per-category toggles, priority floor, quiet hours, weekend pause, vacation mode, snooze, critical-override toggle. Every control has a sane default; the journey is optional, never required (15).

J6 — Trust check (the journey that retains users)

User doubts a claim → clicks the evidence label → sees source URL(s), observation timestamp, and whether the claim is observed, corroborated, or inferred. Doubt resolved in one click or the claim is visibly downgraded.

05 — Functional Requirements

Status	Approved
Owner	Engineering
Depends on	02

Requirement IDs are stable and referenced by tests (24).

FR-100 Onboarding & Company Understanding

- **FR-101** The system SHALL authenticate via email magic link and GitHub OAuth.
- **FR-102** The system SHALL require exactly one input to begin: a company website URL.
- **FR-103** The Company Understanding Engine SHALL extract: industry, business model, target audience, products, features, pricing, technology, keywords, value proposition, customer segment, company size, market positioning.
- **FR-104** All extracted profile fields SHALL be user-editable and re-derivable on demand.

FR-200 Competitor Discovery

- **FR-201** The system SHALL propose competitors automatically without user searching.
- **FR-202** Each candidate SHALL carry a confidence score (internal numeric; UI shows bucket) and human-readable reasoning.
- **FR-203** Users SHALL be able to accept, reject, and manually add competitors.
- **FR-204** Rejections SHALL be persisted and excluded from re-suggestion.

FR-300 Signal Collection

- **FR-301** Collection SHALL occur only from public, non-authenticated sources on the allow-list (27).
- **FR-302** The fetch layer SHALL check robots.txt before every crawl and honor per-domain rate limits (21).
- **FR-303** The system SHALL collect: page/pricing diffs, hiring signals (public ATS JSON + careers pages), blog/release RSS, GitHub public activity, free news/EDGAR business events, model-provider ecosystem announcements.
- **FR-304** The system SHALL backfill historical snapshots via the Wayback Machine CDX API at competitor-add time.
- **FR-305** Every collection run SHALL write a health record (source, status, latency, items) (FR for metric 5 in 01§7).

FR-400 Intelligence Layer

- **FR-401** Every signal SHALL pass the pipeline Raw Event → Context → Business Impact → Strategic Meaning → Recommendation before user display.
- **FR-402** The system SHALL deduplicate signals across sources prior to interpretation.
- **FR-403** The system SHALL assign significance (critical/high/medium/low) per signal, per tracked company context.
- **FR-404** Every stored insight SHALL carry exactly one evidence label: `verified` | `inference` | `recommendation`.
- **FR-406** Every insight SHALL expose its full reasoning chain — Signal → Impact → Reason → Recommendation — as distinct, user-visible fields; a recommendation without a visible Reason step SHALL fail validation.
- **FR-407** Every recommendation SHALL carry an evidence-derived conviction score with its supporting signal families enumerated (e.g., hiring trend + GitHub activity + documentation updates) per ADR-011.
- **FR-405** The system SHALL record user feedback (useful / didn't matter) per insight.

FR-500 Reports

- **FR-501** The system SHALL generate a Baseline Intelligence Report after onboarding (sections per 02§2.3).
- **FR-502** The system SHALL generate Daily, Weekly, Monthly, and Quarterly reports per user schedule.
- **FR-503** Every report section SHALL answer: what happened, what changed, why it matters, what to do.
- **FR-504** Reports SHALL be exportable (HTML → PDF in V1).

FR-600 Search

- **FR-601** The system SHALL answer natural-language queries over stored evidence with labeled claims.
- **FR-602** Search SHALL support entity analysis ("Analyze X") and comparison ("Compare us with Y").

FR-700 Notifications (full spec: 15)

- **FR-701** Channels: email and Telegram in V1; architecture SHALL support adding WhatsApp/Slack/Discord/push without core changes.
- **FR-702** Delivery time SHALL be user-selectable at HH:MM precision, interpreted in the user's IANA timezone with DST handled.
- **FR-703** Frequencies: daily, weekdays-only, weekly, monthly, instant-critical-only.
- **FR-704** Per-category and per-topic preference toggles SHALL be supported (competitor, technology, engineering, hiring, customer intelligence — itemized per 15§4).
- **FR-705** Priority floor filter: critical-only / high+ / medium+ / everything.
- **FR-706** Related events SHALL be grouped into a single smart digest rather than sent individually.
- **FR-707** Quiet hours, weekend pause, vacation mode, and snooze SHALL be supported; critical alerts MAY override quiet hours only if the user opts in.

FR-800 Trust

- **FR-801** No UI surface SHALL render an AI inference with the same visual treatment as verified information (17, 18).
- **FR-802** Every claim SHALL be traceable to source URL(s) and observation timestamp in ≤ 1 interaction.

FR-1000 Decision Workspace (30)

- **FR-1001** Every recommendation and insight SHALL offer: Save as Opportunity, Ignore, Create Feature, Add to Roadmap, Review Later; Export to GitHub Issues SHALL render as a disabled future slot.
- **FR-1002** Decisions SHALL retain full provenance (decision → insight → signals → sources).
- **FR-1003** Roadmap-lite SHALL provide exactly three stages: Considering, Planned, Building.
- **FR-1004** Decision actions SHALL feed the significance/personalization loop (FR-405 extended): saved/created/roadmapped = positive, ignored = negative.
- **FR-1005** A Review Later item SHALL resurface in the CEO Brief on its chosen date.

FR-1100 Market Timeline

- **FR-1101** The system SHALL render a chronological timeline of signals and insights filterable by competitor, category, significance, and date range.
- **FR-1102** Timeline SHALL be answerable via Search ("show me everything that happened during the last 6 months") returning the same view.
- **FR-1103** Backfilled (Wayback) events SHALL appear on the timeline, visually marked as archive-derived.

FR-900 Extensibility

- **FR-901** Every signal category and the LLM layer SHALL sit behind a provider interface; paid providers register via configuration (env key present) and degrade gracefully when absent (08, 10, 28/ADR-003).

06 — Non-Functional Requirements

Status	Approved
Owner	Engineering
Depends on	01\$2 (MVP constraints), 05

NFR-1 Cost

- **NFR-1.1** Total recurring infrastructure cost target: **\$0/month** on free tiers (23). Any exception requires an ADR (28).
- **NFR-1.2** LLM usage SHALL fit within free-tier daily quotas; quota exhaustion degrades gracefully (defer interpretation, never drop signals) (10).

NFR-2 Compliance & Ethics

- **NFR-2.1** robots.txt compliance and domain allow-listing enforced in code (21). Zero tolerance: a source that cannot be collected compliantly is not collected.
- **NFR-2.2** No authentication-gated, paywalled, or CAPTCHA-protected content is ever accessed.
- **NFR-2.3** Per-domain politeness: ≤ 1 request/2s default, exponential backoff, identifying User-Agent with contact URL.

NFR-3 Reliability

- **NFR-3.1** Scheduled collection success rate $\geq 95\%$ per week per source category; failures visible in the health log within one run.
- **NFR-3.2** A dead source SHALL never silently produce "no changes" — it produces a "collection gap" state surfaced in reports.

NFR-4 Performance (budgets in 22)

- **NFR-4.1** Onboarding: URL $\rightarrow$ editable company profile ≤ 2 min; $\rightarrow$ confirmed competitors ≤ 10 min.
- **NFR-4.2** Baseline report ready ≤ 30 min after confirmation.
- **NFR-4.3** Dashboard p95 page load ≤ 2.5 s on free-tier hosting.
- **NFR-4.4** Notification delivery within ± 5 min of scheduled HH:MM.

NFR-5 Data

- **NFR-5.1** Storage budget ≤ 0.5 GB (free Postgres): store hashes, extracted text, structured diffs; raw HTML only for latest snapshot per page.
- **NFR-5.2** All signals immutable once written; corrections are new records (auditability).

NFR-6 Security & Privacy (21)

- **NFR-6.1** Secrets only in environment/secret manager; never in repo.
- **NFR-6.2** User data: email, timezone, preferences, company profile. No collection of user analytics beyond product telemetry; deletable on request.

NFR-7 Maintainability / Portfolio Quality

- **NFR-7.1** A technical reviewer can trace any UI claim $\rightarrow$ insight $\rightarrow$ signal $\rightarrow$ source URL using only the docs and schema.
- **NFR-7.2** Every provider interface has ≥ 1 free implementation and a documented paid slot.
- **NFR-7.3** Code standards per 26; docs updated in the same PR as behavior changes.

NFR-8 Timescale honesty

- The product's cadence is hours-to-days, never seconds. No feature may imply real-time guarantees.

Status	Approved
Owner	Design
Depends on	02, 04

Navigation model

MarketMind	
└ Today	← default view: latest digest, AI strategic summary
└ Competitors	
└ [Competitor page]	← profile, timeline, signals by category, SWOT
└ Compare	← side-by-side (pricing / features / tech / hiring)
└ Intelligence	
└ Technology	← model-provider ecosystem feed (interpreted)
└ Engineering	← GitHub-derived signals
└ Business	← funding/news (labeled partial coverage)
└ Customers	← sentiment slice (labeled narrow coverage)
└ Timeline	← market chronology; filters: competitor / category / range
└ Decisions	
└ Inbox	← undecided recommendations (significance × conviction)
└ Opportunities	
└ Roadmap	← Considering / Planned / Building
└ Decision log	
└ Reports	
└ Baseline	
└ Daily / Weekly / Monthly / Quarterly archive	
└ Export	
└ Search	← conversational analyst
└ Settings	
└ Company profile (editable understanding output)	
└ Competitors (accept/reject/add)	
└ Notifications (channels, schedule, preferences, quiet hours)	
└ Data & sources (allow-list visibility, collection health)	
└ Account	

Content hierarchy rules

- 1. **Conclusion first.** Every page leads with interpretation; raw signals live one level down ("Evidence" tab).
- 2. **Evidence labels are structural**, not decorative — they appear at the claim level, not the page level.
- 3. **Coverage honesty is IA:** categories with partial/narrow coverage carry a permanent, unobtrusive coverage badge linking to 29's user-facing equivalent.
- 4. **Empty states teach.** A category with no compliant source for a given competitor states why ("Competitor has no public repos") rather than showing zero.

URL structure

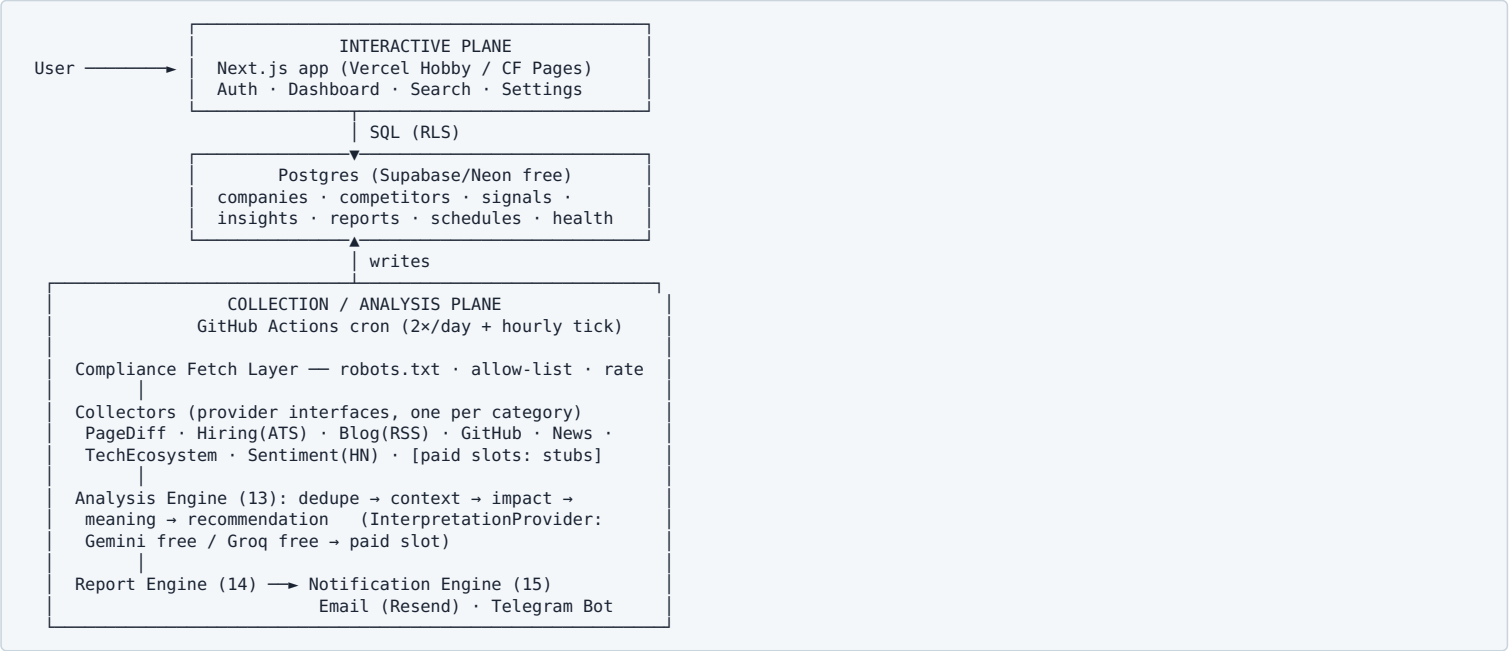
/today
/competitors/:slug
/competitors/compare?a=:slug&b=:slug
/intelligence/:category
/timeline?competitor=&category=&from=&to=
/decisions/:view
/reports/:type/:id
/search?q=
/settings/:section

Status	Approved
Owner	Architecture
Depends on	01\$2, 05, 06

1. Shape

Two planes, strictly separated because free-tier serverless functions have short execution ceilings:

- **Interactive plane** — the web app (auth, dashboard, settings, search, report viewing). Serverless, request/response, never runs long jobs.
- **Collection/analysis plane** — scheduled jobs (crawl, diff, interpret, generate, deliver). Runs on GitHub Actions cron (free for public repos); each run is an auditable log.



2. The provider-interface rule (load-bearing; ADR-003)

One interface **per signal category**, not per vendor:

```
interface SignalProvider {
  category: SignalCategory;
  tier: "free" | "paid";
  isConfigured(): boolean;           // env key present?
  collect(target: Competitor, since: Date): Promise<Signal[]>;
}
```

- Registry resolves providers at runtime: paid provider used **iff configured**, else free provider, else explicit `coverage_gap` record. Graceful degradation is therefore the default execution path, exercised daily — not an untested error branch.
- The same pattern wraps LLMs (`InterpretationProvider`, 10) and delivery channels (`DeliveryChannel`, 15).

3. Data flow contract

`Signal` is the universal unit (schema in 09): every collector normalizes into it; every downstream stage consumes only it. `source_type` and evidence labeling are set at collection time and never inferred later.

4. Scheduling

- **2x/day full collection run** (matches days/weeks product timescale, NFR-8).
- **Hourly tick** for: notification dispatch at users' HH:MM local times (15\$3), critical-alert evaluation, retry queue.
- Timezone math is done in the scheduler (UTC storage, IANA tz conversion per user) — see ADR-006.

5. Failure philosophy

Every stage writes health records (FR-305). A failed source degrades to a visible gap, never a silent "no news." Retries with backoff; poison entries parked with reason.

Status	Approved
Owner	Engineering
Depends on	08

Postgres (Supabase/Neon free tier, ≤ 0.5 GB — NFR-5.1). All timestamps UTC (timestampz). RLS on all user-scoped tables.

Core schema (DDL sketch)

```
-- Identity & workspace
create table users (
  id uuid primary key default gen_random_uuid(),
  email text unique not null,
  tz text not null default 'UTC',          -- IANA, drives 15
  created_at timestampz default now()
);

create table companies (
  id uuid primary key default gen_random_uuid(),
  user_id uuid references users not null,
  url text not null,
  profile jsonb not null,                  -- Understanding Engine output (FR-103), user-editable
  profile_edited_at timestampz
);

create table competitors (
  id uuid primary key default gen_random_uuid(),
  company_id uuid references companies not null,
  url text not null,
  name text not null,
  status text not null check (status in ('suggested','accepted','rejected','manual')),
  discovery_confidence numeric,           -- internal only; UI shows buckets
  discovery_reasoning text,
  created_at timestampz default now()
);

-- Watchable surfaces per competitor (pricing page, blog feed, ATS board, repo...)
create table watch_targets (
  id uuid primary key default gen_random_uuid(),
  competitor_id uuid references competitors not null,
  category text not null,                 -- page_diff|hiring|blog|github|news|tech_ecosystem|sentiment
  locator text not null,                 -- URL / feed / board slug / repo
  provider text not null,                 -- resolved provider id
  active boolean default true
);

-- The universal unit (08§3). Immutable (NFR-5.2).
create table signals (
  id uuid primary key default gen_random_uuid(),
  watch_target_id uuid references watch_targets not null,
  category text not null,
  observed_at timestampz not null,
  source_url text not null,
  source_type text not null check (source_type in ('observed','corroborated','inferred')),
  content_hash text not null,             -- dedupe + LLM-cache key (10)
  payload jsonb not null,                 -- normalized diff / posting / release / article
  backfilled boolean default false,       -- Wayback-derived
  unique (watch_target_id, content_hash)
);

-- Interpreted output of the Analysis Engine (13)
create table insights (
  id uuid primary key default gen_random_uuid(),
  company_id uuid references companies not null,
  evidence_label text not null check (evidence_label in ('verified','inference','recommendation')),
  significance text not null check (significance in ('critical','high','medium','low')),
  title text not null,
  so_what text not null,
  reasoning_chain jsonb,                  -- {signal, impact, reason, recommendation} – FR-406
  conviction jsonb,                       -- {score, factors[]} evidence-derived – ADR-011
  recommendation text,
  signal_ids uuid[] not null,             -- provenance (FR-802)
  model_id text,                          -- which InterpretationProvider produced it
  created_at timestampz default now()
);

create table decisions (
  id uuid primary key default gen_random_uuid(),
  company_id uuid references companies not null,
  insight_id uuid references insights not null,
  action text not null check (action in ('opportunity','ignored','feature','roadmap','review_later')),
  stage text check (stage in ('considering','planned','building')),
  brief text, review_at date, reason text,
  created_at timestampz default now()
);

create table insight_feedback (
  insight_id uuid references insights,
  user_id uuid references users,
  verdict text check (verdict in ('useful','didn't_matter')),
  created_at timestampz default now(),
  primary key (insight_id, user_id)
);
```

```

create table reports (
  id uuid primary key default gen_random_uuid(),
  company_id uuid references companies not null,
  type text check (type in ('baseline','daily','weekly','monthly','quarterly')),
  period_start date, period_end date,
  body jsonb not null,                -- structured sections; rendered by 14/17
  created_at timestamptz default now()
);

-- Notifications (15)
create table notification_settings (
  user_id uuid primary key references users,
  channels jsonb not null default '{"email":true,"telegram":false}',
  send_time time not null default '08:00', -- HH:MM, interpreted in users.tz
  frequency text not null default 'daily'
  check (frequency in ('daily','weekdays','weekly','monthly','instant_critical')),
  weekly_day int, monthly_day int,
  priority_floor text not null default 'medium'
  check (priority_floor in ('critical','high','medium','all')),
  category_prefs jsonb not null default '{}', -- per-topic toggles (15$4)
  quiet_hours jsonb,                        -- {"start":"22:00","end":"07:00"}
  weekend_pause boolean default false,
  vacation_until date,
  snooze_until timestamptz,
  critical_overrides_quiet boolean default false
);

create table deliveries (
  id uuid primary key default gen_random_uuid(),
  user_id uuid references users,
  channel text, kind text, report_id uuid,
  scheduled_for timestamptz, sent_at timestamptz,
  status text check (status in ('sent','failed','suppressed_quiet','suppressed_pref'))
);

create table collection_health (
  id bigint generated always as identity primary key,
  run_id text not null, watch_target_id uuid,
  status text check (status in ('ok','empty','blocked','error','gap')),
  detail text, latency_ms int, items int,
  created_at timestamptz default now()
);

create table llm_cache (
  cache_key text primary key,
  output jsonb not null,
  created_at timestamptz default now()
);

```

Storage discipline

- Raw HTML: latest snapshot only, in object storage if needed; DB stores extracted text + structured diff.
- `signals.payload` capped (8 KB typical) — extraction happens at collection, not later.
- Indexes: `signals(watch_target_id, observed_at desc)`, `insights(company_id, created_at desc)`, `deliveries(user_id, scheduled_for)`.

Status	Approved
Owner	AI Engineering
Depends on	08, 09; ADR-003/004/005 (28)

1. Provider abstraction

```
interface InterpretationProvider {
  id: string; // "gemini-free" | "groq-free" | "claude-paid"...
  tier: "free" | "paid";
  limits: { rpm: number; rpd: number; tpm?: number };
  isConfigured(): boolean;
  interpret(batch: SignalBatch, task: TaskSpec): Promise<Interpretation>;
}
```

V1 cascade: gemini-free → groq-free → defer. Paid slot (claude-paid / gpt-paid) activates when its key exists — zero code change (FR-901). Failover between the two free providers is implemented and demoed (portfolio requirement, NFR-7.2).

2. Quota-shaped design (free tiers make RPM/RPD first-class inputs)

- 1. **Batch, don't stream:** one run interprets all of a competitor's new signals in a few structured prompts, never one call per signal.
- 2. **Cache on content:** llm_cache keyed on (model, prompt_version, content-hash set). Unchanged content is **never** reinterpreted (09).
- 3. **Budgeter:** each run gets a token/request budget; ordering = significance-pre-score desc; leftovers defer to the next run with status deferred, never dropped (NFR-1.2).

3. Task taxonomy (each has a versioned prompt template)

Task	Input	Output
understand_company	site text	company profile JSON (FR-103)
discover_competitors	profile + search snippets	candidates + reasoning + confidence
presignificance	raw signal	cheap 0–1 score (small/free model) — gatekeeper before full interpretation
interpret	deduped signal batch + both-sides company context	per-signal: context, business impact, strategic meaning, significance, evidence label, recommendation
digest	period's insights	grouped digest + AI Strategic Summary
report_section	insights + history	report section JSON (14)
search_answer	question + retrieved evidence	labeled, cited answer (16)

4. The interpretation contract (the heart — Manifesto §"Philosophy of Intelligence")

Prompts require structured output answering, for every event: *what changed · why it matters · who is affected · how it affects THIS user's company · should the founder care · what to do*. The user's own company profile (from the identical pipeline) is always in context, so recommendations are grounded in **public knowledge of both sides** — never in private data the system doesn't have (ADR-002).

Hard rules enforced by output schema + validator:

- Every claim tagged `verified | inference | recommendation`; verified claims must reference `signal_ids`.
- No numeric confidence in user-facing text — **evidence buckets only:** *observed / corroborated by N sources / single-source inference* (ADR-004: LLM numeric confidence is not calibratable; false precision destroys trust).
- Refuse-to-guess: if evidence is insufficient, output `insufficient_evidence`, which renders as an honest gap, not filler.

5. Quality loop

- `insight_feedback` verdicts adjust per-category significance thresholds (simple Bayesian update in V1; learned model is V3).
- Weekly automated audit: sample 10 insights → verify each `verified` claim resolves to its source → mismatches fail CI-style and page the maintainer.
- Prompt templates are versioned files in-repo; every insight records `model_id` + prompt version for reproducibility.

6. What the AI layer must never do

Fabricate sources; upgrade an inference to verified; interpret content the compliance layer didn't approve; exceed provider ToS; present a recommendation without its evidence chain.

11 — Company Understanding & Competitor Discovery Engine

Status	Approved
Owner	AI Engineering
Depends on	10, 12

Quality bar (stated up front, per Vision §5 Moment 1)

- Editable company profile ≤ 2 min after URL submission.
- Confirmed competitor set ≤ 10 min.
- Suggestion acceptance target ≥ 70% for well-documented SaaS; below 50% in testing blocks release.

Stage A — Company Understanding Engine

1. Compliance-fetch the user's site: home, pricing, product, about, docs index, careers, blog feed (same collectors as competitors — symmetry is deliberate, ADR-002).
2. `understand_company` task (10§3) produces the structured profile (industry, business model, audience, products, features, pricing, technology, keywords, value proposition, segment, size, positioning).
3. Render profile for review; every field editable (FR-104). Edits are ground truth thereafter.

Stage B — Candidate generation (quota-aware)

Signals combined (each cheap, free, compliant): - **Search:** Google Programmable Search JSON API — 100 free queries/day; discovery uses ≤ 15 per onboarding (e.g., "`<category>`" alternatives, "`vs <company>`", `<value-prop keywords>` tool). Quota shaping per ADR-005. - **Comparison-page mining:** competitors named on "vs/alternatives" pages found in search results. - **Ecosystem adjacency:** shared directories/categories on public listing pages; GitHub topic adjacency where relevant. - **AI reasoning:** `discover_competitors` task merges evidence → candidates with reasoning + confidence.

Stage C — Confirmation UX (the human is the analyst's editor)

- Ranked candidates, each with one-line reasoning ("Their pricing page targets the same SMB segment; named on 3 comparison pages").
- Confidence shown as buckets. Honest framing on thin evidence: "2 confident matches, 5 guesses — help us calibrate."
- Accept / reject / add-by-URL. Rejections persist (FR-204). Added competitors run through Stage A understanding automatically.

Fallback behavior

- Long-tail/novel category with < 3 confident candidates → the system says so and leads with manual-add, rather than padding with noise. An honest short list beats a confident wrong one (Manifesto belief 2).

Metrics

Acceptance rate, time-to-confirmation, rejection reasons (sampled), and post-hoc "missed competitor" additions in week 1 (a proxy for recall).

12 — Signal Collection Engine

Status	Approved
Owner	Engineering
Depends on	08, 09, 21, 27

1. Compliance Fetch Layer (wraps every request; no bypass path exists)

```
request → allow-list check (27) → robots.txt fetch+cache (24h) → rule evaluation
      → per-domain rate limiter (≥2s spacing, jitter) → fetch (UA with contact URL)
      → on 403/429: exponential backoff, then mark BLOCKED (health record) — never retry-hammer, never spoof
```

Blocked/removed sources become visible `coverage_gap` states (NFR-3.2). This layer is a portfolio showpiece: it logs every decision it makes.

2. Collectors (each implements `SignalProvider`, 08§2)

2.1 PageDiffCollector — grade A

- Watch targets: pricing, home, product, changelog pages (seeded from sitemap.xml + understanding stage).
- Fetch → extract main text (strip nav/footer) → hash → structured diff vs. last snapshot (added/removed/changed blocks; price tokens parsed specially).
- **Wayback backfill:** at competitor-add, query CDX API for historical snapshots of pricing/home (up to 24 months, ≤ 12 snapshots); diff chronologically; write `backfilled=true` signals. This gives the Baseline Report longitudinal facts on day one (ADR-001).

2.2 HiringCollector — grade A

- Detect ATS from careers page: **Greenhouse / Lever / Ashby public JSON board endpoints** (documented, unauthenticated, intended for public consumption) → structured postings (title, team, location, posted_at).
- Fallback: careers-page diff via 2.1.
- Derived series: openings by function over time; new-role events; senior-hire events.

2.3 ContentCollector — grade A

- RSS/Atom for blog, changelog, release notes. Feed discovery via `<link rel=alternate>` + common paths. New-entry signals with extracted text.

2.4 GitHubCollector — grade B

- REST API, token, 5k req/hr free. Releases, tags, repo creation, stars/forks deltas, top-level dependency-file changes on public repos. Absent public repos → explicit "no public engineering signal" state.

2.5 NewsCollector — grade B (labeled partial)

- Google News RSS per competitor name; competitor press pages; **SEC EDGAR full-text search** for Form D/S-1 (US). All outputs labeled; coverage badge "partial" in UI. Paid slot: Crunchbase provider (stub).

2.6 TechEcosystemCollector — grade A

- Official blogs/changelogs/status feeds of model providers (OpenAI, Google/Gemini, Anthropic, Groq, Cerebras, OpenRouter, DeepSeek, Mistral, Meta/Llama, xAI, Perplexity — list in 27). Tracks models, pricing, context windows, SDK/API changes, MCP/function-calling/memory/voice/vision/embeddings announcements, deprecations. These are global watch targets shared across all users (collected once).

2.7 SentimentCollector — grade C (narrow slice, labeled)

- **HN via Algolia API** (fully open). Reddit only within current free-tier API terms, low rate. **G2/Capterra: excluded — ToS prohibits scraping;** `ReviewSignalProvider` exists as a visible stub ("requires licensed data partner").

2.8 Paid stubs (V1 ships the interface, not the vendor)

`FirmographicsProvider`, `ReviewSignalProvider`, `FundingProvider(paid)`, `TechStackProvider(paid)` — each `isConfigured()` `=== false` in V1 and renders as an honest locked slot in Settings → Data & sources.

3. Normalization

Every collector emits `Signal` rows (09) with `source_url`, `observed_at`, `source_type='observed'`, `content_hash`, normalized `payload`. Dedup at insert (`unique(watch_target_id, content_hash)`) and again cross-source in 13.

4. Scheduling & budget

2x/day full run (08§4). Per-run fetch budget ≈ 40 requests/competitor; ordering: pricing > careers/ATS > blog > GitHub > news > sentiment.

13 — Analysis Engine (the Intelligence Layer)

Status	Approved
Owner	AI Engineering
Depends on	10, 12

The pipeline (FR-401)

Raw Event → Context → Business Impact → Strategic Meaning → Recommendation

Raw data is never shown first; it survives only as the Evidence view behind each insight.

Stage 1 — Dedupe & cluster

Cross-source dedup (same funding round from 3 outlets = one event). Clustering: signals about the same underlying move (price cut + landing-page reframe + sales hiring) merge into one **event cluster** — clusters are where the best insights live.

Stage 2 — Pre-significance gate

Cheap presignificance scoring (10\$3) filters noise (typo-level page diffs, boilerplate posts) **before** spending interpretation quota. Threshold per category, adjusted by feedback (Stage 5).

Stage 3 — Interpretation

interpret task with both-sides context (user profile + competitor profile + relevant history). Output per event/cluster: - **What changed** (verified claims → signal_ids) - **Why it matters / who is affected** (inference) - **Impact on this user's company** (inference, grounded in the user's public profile only — ADR-002) - **Significance:** critical / high / medium / low (drives 15's priority floor) - **Reasoning chain (FR-406):** the output is structured as Signal → Impact → **Reason** → Recommendation, and the Reason step is rendered to the user — users trust reasoning, not conclusions. A recommendation whose Reason is empty fails validation (Stage 4). - **Conviction (FR-407 / ADR-011):** each recommendation carries an evidence-derived conviction score computed deterministically from the breadth and agreement of supporting signal *families* (e.g., hiring trend + GitHub activity + documentation updates ⇒ high). The factors are always listed alongside the number — the score is explainable arithmetic over evidence, never the model's self-reported confidence. - **Recommendation** (labeled recommendation; advisory always — Manifesto non-goal) - Or insufficient_evidence → rendered as an honest gap.

Never "OpenAI released a new model." Always: what changed, why it matters, who's affected, how it affects *you*, should you care, what to do.

Stage 4 — Validation

Schema validation; every verified claim must resolve to signal_ids; evidence-label sanity (a claim citing zero signals cannot be verified); profanity/PII scrub. Failures re-queue once, then park with reason.

Stage 5 — Feedback loop

👍/👎 per insight (FR-405) and — much stronger — Decision Workspace actions (FR-1004): saved/created/roadmapped insights raise the weight of their pattern; ignored insights lower it. "Didn't matter"/ignored x3 on the same pattern suppresses it for that user.

Personal Intelligence (V2/V3 direction, 25): the same decision/feedback stream learns per-founder attention — e.g., ignores hiring, loves pricing, always reads OpenAI updates — and reorders digests and Brief priorities accordingly. V1 collects the data; V2 personalizes ranking; V3 learns models per segment.

Significance rubric (V1 defaults)

Level	Examples
Critical	Direct-competitor pricing change; competitor launches feature matching user's core value prop; acquisition of/by a competitor
High	Major model-provider pricing/capability change relevant to user's stack; competitor senior hire in user's domain; funding event
Medium	Notable release cadence shift; sustained hiring in one function; significant content pivot
Low	Routine posts, minor page edits that pass the gate

14 — Report Generation Engine

Status	Approved
Owner	AI Engineering, Design
Depends on	13, 15, 17

Universal report contract (FR-503)

Every report and every section answers, in order: **What happened** → **What changed** → **Why it matters** → **What should I do**. Every claim carries its evidence label; every verified claim links to sources.

Report types

Baseline Intelligence Report (once, post-onboarding; ≤ 30 min)

A **snapshot**, not a change report — solves cold start. Sections: market overview · competitor overview · technology comparison · pricing comparison · feature comparison · AI adoption · engineering activity · hiring trends · customer sentiment (coverage-labeled) · SWOT per competitor · threat level · opportunity areas · missing features · strategic recommendations. Longitudinal depth comes from Wayback backfill ("Competitor X changed pricing 3× in 12 months") — the report's signature moment.

Daily CEO Brief (upgraded Daily Digest)

Delivered per user schedule (15). The most engagement-critical artifact in the product — designed to be the first thing a founder reads each morning:

```
Good Morning, <name>.

Yesterday:
✓ OpenAI released new Responses API
✓ Competitor reduced pricing
✓ Two competitors hired ML Engineers
✓ Customer sentiment shifted positive

Today's Recommendation
Prioritize Voice AI integration.

Why: competitor hiring trend + GitHub activity + documentation updates
all point the same direction.           ← the Reason step, always visible

Market urgency: High
Conviction: High (3 independent signal families)  ← factors enumerated, ADR-011
[ Save as Opportunity ] [ Add to Roadmap ] [ Ignore ] [ Review Later ]
```

Structure: greeting → yesterday's confirmed events (checklist, evidence-labeled) → exactly **one** prioritized Today's Recommendation with reasoning chain, urgency, conviction + factors, and inline decision actions (30) → remaining grouped events → deferred/low items collapsed. Empty day = one honest line, not padding. Review-Later items due today resurface here (FR-1005).

Weekly Strategic Report

Built for forwarding (persona P2): week's clusters ranked by significance; per-competitor movement; technology-ecosystem shifts affecting the user's stack; 2–4 recommended actions with evidence chains; coverage/health appendix (gaps disclosed — NFR-3.2).

Monthly Market Report

Trends over weeks: hiring trajectories, pricing evolution, release cadence, sentiment slice trend. Includes "what we said → what happened" retrospective on prior recommendations (trust feature).

Quarterly Competitive Review

Landscape-level: positioning shifts, SWOT deltas vs. baseline, threat-level changes, strategic themes.

Generation mechanics

`report_section` tasks (10§3) fill a typed JSON structure stored in `reports.body` (09); renderer (17) is deterministic from that JSON. Export: HTML → PDF (V1). All reports archived and diffable.

15 — Intelligent Notification & Delivery System

Status	Approved
Owner	Engineering
Depends on	08\$4, 09 (notification_settings, deliveries), 14

MarketMind AI never forces users into a fixed schedule. Users have complete control over **when, how, and what** they receive — layered on good defaults so control is optional.

1. Delivery channels

Channel	Status
Email (Resend free tier)	MVP
Telegram (Bot API — free)	MVP
WhatsApp	Future
Slack	Future
Discord	Future
Mobile push	Future

Multiple channels may be enabled simultaneously. Each channel implements:

```
interface DeliveryChannel {
  id: "email" | "telegram" | "whatsapp" | "slack" | "discord" | "push";
  isConfigured(user): boolean;
  render(digest: Digest): ChannelPayload;    // channel-appropriate formatting
  send(payload): Promise<DeliveryResult>;
}
```

Future channels are new implementations only — no core changes (FR-701).

2. Custom delivery time — HH:MM precision

- Any time of day: 05:30, 08:00, 11:45, 13:15, 16:30, 21:00, 23:59 — a full time picker, not presets.
- Stored as `send_time` (HH:MM) + `users.tz` (IANA). **All scheduling math: store UTC, convert per-user at evaluation time.** DST is handled by the IANA tz database — a user set to 08:00 Europe/Berlin gets 08:00 wall-clock year-round (ADR-006).
- Dispatcher: hourly tick (08\$4) evaluates, each run, which users' local HH:MM falls in the window; enqueues sends. Delivery SLA ±5 min (NFR-4.4).

3. Frequency

`daily` · `weekdays only` · `weekly` (chosen weekday) · `monthly` (chosen day, clamped for short months) · `instant alerts for critical events only`. Fully custom cron-like schedules: future enhancement (25).

4. Notification preferences — per-topic toggles (`category_prefs` jsonb)

Competitor Intelligence: new feature launches · pricing changes · product updates · marketing campaigns · new partnerships · funding announcements · acquisitions. **Technology Intelligence:** per-provider toggles — OpenAI · Gemini · Claude · Groq · Cerebras · OpenRouter · DeepSeek · Mistral · Llama · xAI (list maintained in 27). **Engineering Intelligence:** GitHub releases · new repositories · major commits · framework migrations · API changes · SDK releases. **Hiring Intelligence:** AI engineer · backend · frontend · product · leadership hiring. **Customer Intelligence:** negative sentiment spikes · positive trends · top feature requests · new reviews (coverage-labeled).

Suppressed items are recorded as `suppressed_pref` in `deliveries` (auditable, reversible).

5. Priority floor

`Critical only` · `High + Critical` · `Medium and above` · `Everything`. Applied after preferences; significance comes from 13's rubric.

6. Smart Digest — delivered as the Daily CEO Brief (FR-706, 14)

Never 50 separate notifications. The `digest` task (10\$3) groups related events into one delivery:

```
Morning Intelligence Report — Today's Summary
• 3 competitors released new features
• 2 pricing changes detected
• OpenAI announced a new API
• Gemini updated model pricing
• One competitor hired 5 AI engineers

AI Strategic Summary
The market is accelerating investment in AI agents. Two direct competitors are
expanding their AI engineering teams. Consider prioritizing agent capabilities
in your roadmap during the next sprint.
```

Instant-critical alerts bypass grouping but still deduplicate within a 30-min window.

7. Snooze & Quiet Hours

- Quiet hours** (e.g., 22:00–07:00 local): non-critical sends are shifted to quiet-hours end, not dropped (`suppressed_quiet` → rescheduled).
- Weekend pause** · **Vacation mode** (until date) · **Temporary snooze** (hours/days).
- Critical override:** critical alerts may break quiet hours **only if** the user enables `critical_overrides_quiet` (default off).

8. Anti-spam invariants

Max 1 scheduled digest per frequency period per channel; instant alerts rate-limited ($\leq 3/\text{day}$, excess folds into next digest); every send audited in `deliveries`; one-click per-channel disable in every message footer.

16 — Search Engine (Conversational Analyst)

Status	Approved
Owner	AI Engineering
Depends on	09, 10, 13

Behavior contract

Search behaves like talking to a senior product strategist, answering over **collected evidence** — not open web guessing. Supported intents:

Intent	Example	Handling
Entity analysis	"Analyze Notion."	If tracked: synthesize stored profile + signals + insights. If untracked: one-off understanding run within quota, clearly labeled "one-time analysis, not monitored."
Comparison	"Compare us with Linear."	Side-by-side over pricing/features/tech/hiring evidence; gaps stated, not filled.
Temporal	"What pricing changes happened this month?"	SQL-backed retrieval over <code>signals</code> (category=page_diff, price tokens) → summarized with citations.
Timeline	"Show me everything that happened during the last 6 months."	Renders the Market Timeline view (FR-1102), month-grouped, filter chips pre-applied.
Trend	"What technologies are competitors adopting?" / "What AI models are becoming popular?"	Aggregation over engineering + tech-ecosystem signals.

Mechanics (V1 — deliberately simple)

- Intent + entity parse** (small model call).
- Retrieval:** structured SQL first (signals/insights are already normalized — most questions are queries, not vector search); Postgres FTS (`tsvector`) over payload text for the rest. `pgvector` is a V2 upgrade, noted in 25 — not needed for MVP scale.
- Synthesis:** `search_answer` task; every claim labeled; verified claims cite signal sources; insufficient evidence stated plainly.

Guardrails

- Answers never exceed the evidence: no fabricated metrics, no invented sources (10§6).
- Untracked-entity analyses consume the shared discovery/search quota (ADR-005) and say so if quota is exhausted ("try after 00:00 UTC").
- Every answer offers "add to monitoring" when the entity isn't tracked.

Status	Approved
Owner	Design
Depends on	07, 13, 18

Design stance — a command center, not an admin dashboard

Bloomberg Terminal meets Linear. One dense, keyboard-driven workspace — not stacked cards. Calm, editorial, text-first typography carries the judgment; information density is earned through hierarchy, not chrome. Concretely: - **⌘K command palette** is the primary navigation ("compare with Linear", "timeline last 6 months", "open decisions inbox"). - **Three-pane workspace** on desktop: navigation rail · primary stream (Today / Timeline / competitor) · context pane (evidence, reasoning chain, decision actions) — the context pane replaces modal-hopping. - Card grids are banned as a default layout; lists, tables, and timelines with strong typographic rhythm instead. - Charts appear only where a trend *is* the point (hiring over time, release cadence).

The evidence-label visual language (FR-801 — the core UI invariant)

Label	Treatment
Verified	Solid badge, strong ink, source-count chip; click → sources + timestamps (≤ 1 interaction, FR-802)
AI Inference	Outlined badge, muted tone, "inferred" prefix; hover explains basis
Recommendation	Distinct accent block, always visually downstream of its evidence; advisory phrasing enforced ("Consider...", never "Do...")

An inference must never carry the visual weight of a verified fact — reviewed in every design PR.

Onboarding: watching the AI work

The understanding/discovery run streams its reasoning like Perplexity/ChatGPT — a live step sequence, not a spinner:

Reading your website...

Understanding your business...

Finding competitors...

Analyzing market...

Generating SWOT...

Preparing your baseline report...

✓

✓ (profile fields appear as extracted)

● "found 4 candidates via comparison pages"

Each step surfaces a one-line genuine artifact of what was found (never fake progress). People trust — and enjoy — watching the analyst think; the stream doubles as the first demonstration of the evidence-first culture.

Key screens

- **Today:** the CEO Brief rendered as the lead block (greeting, yesterday checklist, one prioritized recommendation with reasoning chain + conviction factors + inline `<DecisionBar>`); grouped insight stream below. Every insight expands to its reasoning chain (Signal → Impact → Reason → Recommendation) and raw Evidence. Feedback /🗑 everywhere.
- **Timeline:** month-grouped vertical chronology of the whole market; filter chips (competitor / category / significance / range); backfilled events marked with an archive glyph; any entry opens in the context pane; reachable from Search ("last 6 months") and ⌘K.
- **Decisions:** Inbox (ranked by significance × conviction), Opportunities, Roadmap-lite three-column board, Decision log — provenance one click away on every card.
- **Competitor page:** profile header; vertical timeline of insights; category tabs (Pricing / Hiring / Product / Engineering / News / Sentiment) each with coverage badge; SWOT panel labeled `inference`.
- **Compare:** two-column table; rows only where evidence exists on both sides; gaps say "no compliant public signal."
- **Reports:** rendered from typed JSON (14); print/PDF-clean stylesheet.
- **Search:** single input; streamed answer with inline labels/citations; suggested follow-ups.
- **Settings → Notifications:** channel cards, HH:MM picker with tz shown, frequency, per-topic toggle groups (15\$4), priority floor, quiet-hours range slider, weekend/vacation/snooze, critical-override switch.
- **Settings → Data & sources:** allow-list view, per-source health (green/gap/blocked), and **locked paid-provider slots** displayed honestly ("Firmographics — requires data partner").

States that must be designed (not defaulted)

Empty-with-reason (coverage gap), quota-deferred ("interpretation queued — free-tier quota resets 00:00 UTC"), collection-blocked, insufficient-evidence, vacation-mode banner.

Accessibility

WCAG 2.1 AA; labels never color-only (badge shape + text); full keyboard nav; reduced-motion respected.

Status	Approved
Owner	Design
Depends on	17

Tokens

```
:root {
  /* Ink & surface (light) */
  --mm-ink: #16212e;      --mm-ink-muted: #5b6773;
  --mm-surface: #ffffff;   --mm-surface-alt: #f5f7f9;
  --mm-border: #e3e8ee;

  /* Brand */
  --mm-accent: #0e5a8a;    --mm-accent-soft: #e8f1f7;

  /* Evidence labels (the semantic core) */
  --mm-verified: #1e6b3a;  --mm-verified-bg: #e9f5ee;
  --mm-inference: #6b5d1e; --mm-inference-bg: #f7f3e3;
  --mm-recommend: #4a2e8a; --mm-recommend-bg: #efe9f9;

  /* Significance */
  --mm-critical: #a3241a;   --mm-high: #b8641a;
  --mm-medium: #0e5a8a;    --mm-low: #5b6773;

  /* Type */
  --mm-font-ui: Inter, system-ui, sans-serif;
  --mm-font-mono: "JetBrains Mono", ui-monospace, monospace;

  /* Scale (rem): 12 13.5 15 17 20 24 30 38 */
  --mm-radius: 8px;  --mm-radius-lg: 12px;
  --mm-space: 4px;   /* 4-pt spacing system: 4/8/12/16/24/32/48 */
}
```

Dark theme: inverted ink/surface pairs; evidence hues keep ≥ 4.5:1 contrast on both themes.

Rules

- Evidence-label colors are **reserved** — never reused decoratively.
- Significance is a dot/left-border, not a filled card (calm > alarm; critical is the only saturated treatment).
- One accent. No gradients in product UI. Charts use ink + accent + evidence hues only.
- Type hierarchy carries structure: reports use the 20/17/15 sizes with generous leading (1.55) — dashboards must read like well-set documents.
- Motion: 150–200ms ease-out, opacity/transform only.

19 — Component Library

Status	Approved
Owner	Frontend
Depends on	17, 18

Stack: React + TypeScript, Tailwind mapped to 18's tokens, Radix primitives for a11y. Storybook is the contract; every component ships with states + a11y notes.

Semantic components (product-specific)

- `<EvidenceBadge label="verified|inference|recommendation" sources={n} />` — the most important component in the product; encodes 17's visual language; click opens `<SourcePopover>`.
- `<InsightCard>` — title, so-what, EvidenceBadge, SignificanceDot, feedback buttons, Evidence expander.
- `<StrategicSummary>` — the digest's lead paragraph treatment.
- `<CoverageBadge state="full|partial|narrow|gap|blocked" reason />`
- `<CompetitorTimeline>` / `<CompareTable>` (renders only both-sided evidence rows).
- `<ReportRenderer body={ReportJSON}>` — deterministic renderer for 14's typed sections.
- `<ProviderSlot configured={bool}>` — honest locked-slot card for paid providers.
- `<HealthIndicator>` — per-source collection health.
- `<DecisionBar>` — the six decision actions (GitHub export rendered disabled); one-click, optimistic (30\$1).
- `<ReasoningChain>` — Signal → Impact → Reason → Recommendation, rendered as a vertical chain with evidence labels per step (FR-406).
- `<ConvictionMeter score factors[]>` — number + enumerated supporting signal families; never shown without factors (ADR-011).
- `<MarketTimeline>` — month-grouped chronology; virtualized; archive-glyph for backfilled events.
- `<RoadmapBoard>` — Considering / Planned / Building, drag between stages.
- `<CommandPalette>` (⌘K) — navigation + query launcher.
- `<ThinkingStream>` — onboarding live-step renderer (17).

Notification-settings components

- `<TimePicker24 value="HH:MM" tzLabel />` (full 24h precision, FR-702)
- `<FrequencySelect>` · `<TopicToggleGroup categories={15$4}>` · `<PriorityFloor>`
- `<QuietHoursRange>` · `<PauseControls weekend vacation snooze>` · `<CriticalOverrideSwitch>`
- `<ChannelCard channel state>` (connect/test/disable; Telegram shows bot-link flow)

Primitives

Button, Input, Select, Switch, Tabs, Dialog, Popover, Tooltip, Toast, Skeleton, EmptyState(reason), Table, Badge, Card — all consuming tokens only; no hard-coded colors (lint-enforced, 26).

Status	Approved
Owner	Engineering
Depends on	09, 05

Internal REST API (app ↔ backend). JSON; auth via session (Supabase Auth JWT); all responses carry evidence labeling — the schema enforces the trust contract.

Conventions

- Base: /api/v1 · Errors: { "error": { "code", "message" } } · Pagination: ?cursor=&limit= · Timestamps: ISO-8601 UTC.

Endpoints

POST /companies	{ url }	→ 202 { company_id, understanding_job }
GET /companies/:id		→ profile (+ coverage summary)
PATCH /companies/:id/profile	{ field edits }	→ updated profile (FR-104)
GET /companies/:id/competitors		→ suggested/accepted/rejected lists
POST /companies/:id/competitors	{ url }	→ manual add (runs understanding)
PATCH /competitors/:id	{ status: accepted rejected }	
GET /companies/:id/insights	?significance=&category=&cursor=	
GET /insights/:id		→ full insight + signal provenance
POST /insights/:id/feedback	{ verdict: useful didnt_matter }	
GET /insights/:id/evidence		→ signals[] with source_url, observed_at (FR-802)
GET /companies/:id/reports	?type=	
GET /reports/:id		→ typed body JSON
GET /reports/:id/export	?format=pdf	
POST /search	{ q }	→ { answer_blocks[]: {text, label, signal_ids[]} }
GET /me/notification-settings		
PUT /me/notification-settings	{ channels, send_time:"HH:MM", frequency, weekly_day?, monthly_day?, priority_floor, category_prefs, quiet_hours?, weekend_pause, vacation_until?, snooze_until?, critical_overrides_quiet }	
POST /me/channels/telegram/link		→ { deep_link } (bot linking)
POST /me/channels/:id/test		→ sends test digest
POST /insights/:id/decisions	{ action, stage?, review_at?, reason? }	
GET /companies/:id/decisions	?action=&stage=&cursor=	
PATCH /decisions/:id	{ stage brief review_at }	
GET /companies/:id/decisions/log		
GET /companies/:id/timeline	?competitor=&category=&from=&to=&cursor=	
GET /companies/:id/health		→ per-source collection health (NFR-3)

Response invariants

1. Any text block that makes a claim carries label: verified|inference|recommendation.
2. verified blocks include non-empty signal_ids.
3. Coverage-limited categories include coverage: full|partial|narrow|gap in list responses.

Machine-readable OpenAPI 3.1 file lives at /api/openapi.json and is generated from route schemas (single source of truth; drift fails CI — 24).

Status	Approved
Owner	Engineering
Depends on	06, 12

1. Collection compliance (the product's defining security posture)

- **Allow-list first:** only domains/endpoints cataloged in 27 are fetchable. Adding a source = PR updating 27 with a legal-basis row.
- **robots.txt enforced in code** (cached 24h, fail-closed on parse errors), identifying User-Agent MarketMindBot/1.0 (+contact URL), per-domain rate limits, no login/paywall/CAPTCHA circumvention — ever (Manifesto belief 5; NFR-2).
- 403/429 → backoff → blocked state; we never rotate IPs/UAs to evade blocks.
- Every fetch decision logged (collection_health) — auditable compliance, not asserted compliance.

2. Application security

- Auth: Supabase Auth (magic link, GitHub OAuth); short-lived JWTs; RLS on every user-scoped table (09).
- Secrets: platform secret stores (Vercel/GitHub Actions encrypted secrets); none in repo; quarterly rotation.
- Input handling: fetched web content is untrusted — sanitized before storage, escaped on render, and **treated as data, not instructions, in prompts** (prompt-injection defense: LLM tasks wrap scraped text in delimited data blocks; instructions live only in versioned templates; outputs schema-validated — 10§4).
- Standard headers: CSP, HSTS, X-Content-Type-Options; CSRF on state-changing routes.

3. Data protection & privacy

- Personal data held: email, tz, preferences, company profile. Export + delete on request (full cascade).
- Collected signals are public-web observations of companies; no personal-data enrichment of individuals (hiring signals aggregate roles, never named candidates/employees).
- Telegram linking uses one-time deep-link tokens; chat IDs stored hashed.

4. LLM-specific

- Free-tier providers receive only public collected text + the user's own public profile — never credentials or other users' data.
- Per-task output validators reject label violations (an unverifiable "verified" claim is a security bug, not a quality bug).

5. Threat notes (V1-proportional)

No multi-tenant orgs yet → the main risks are scraped-content injection (mitigated above), SSRF via user-supplied URLs (fetch layer blocks private IP ranges/localhost/link-local; DNS-rebinding check), and quota abuse (per-user onboarding quotas).

22 — Performance & Observability

Status	Approved
Owner	Engineering
Depends on	06

Budgets (free-tier realistic)

Surface	Budget
Dashboard p95 TTI	≤ 2.5 s
Insight/report page p95	≤ 1.5 s (server-rendered from stored JSON)
Search answer first token	≤ 4 s; complete ≤ 15 s
Onboarding: profile ready	≤ 2 min
Baseline report	≤ 30 min
Notification delivery	scheduled HH:MM ± 5 min
Full collection run (10 competitors)	≤ 25 min within Actions limits

Principles

- Precompute, don't compute on request.** Reports and digests are generated in the collection plane and stored; the interactive plane renders stored JSON. LLM calls never happen inside a page request (except Search, which streams).
- Quota is a performance dimension.** The budgeter (10§2) is instrumented like latency: quota-used, deferrals, cache-hit-rate are first-class metrics.
- DB discipline:** indexed access paths listed in 09; no unbounded scans; payload caps keep rows small.

Observability (free-tier)

- `collection_health` is the primary ops surface (per-source status, latency, items, gaps) — rendered in Settings → Data & sources and in the weekly report appendix.
- GitHub Actions run logs = collection audit trail (public, reviewable — a portfolio feature).
- Web vitals via hosting platform analytics (free tier).
- Weekly automated summary: success rate per category, LLM cache-hit %, deferral count, delivery SLA hits — posted as a repo issue.

Alerting (V1-proportional)

Actions job failure → GitHub notification + issue. Two consecutive `blocked/gap` states on the same source → flagged in next digest's health appendix (churn-vs-gap metric, 01§7).

23 — Deployment & Environments

Status	Approved
Owner	Engineering
Depends on	08

Reference free-tier stack (verify limits at build time; swaps allowed within ADR-003's interfaces)

Layer	Choice	Free-tier note
Web app	Vercel Hobby (or Cloudflare Pages)	Serverless; short function ceilings → no jobs in requests
DB + Auth	Supabase free (or Neon + Auth.js)	~0.5 GB Postgres, RLS, Auth included
Scheduler / jobs	GitHub Actions cron (public repo)	Free minutes; runs are auditable logs
LLM	Gemini free API + Groq free	Behind InterpretationProvider; keys in Actions secrets
Email	Resend free	Thousands/month — far beyond MVP needs
Telegram	Bot API	Free
Object storage (latest raw snapshots)	Supabase Storage free	Optional; DB-only is acceptable at MVP scale

Environments

- **prod** — main branch. **preview** — per-PR (Vercel previews) against a branch DB (Neon branching / Supabase preview). No long-lived staging in V1.

Pipeline (GitHub Actions)

1. `ci.yml` — lint, typecheck, unit + contract tests (24), OpenAPI drift check, migration dry-run.
2. `deploy` — Vercel auto-deploy on main after CI green; migrations applied via `migrate` job first (expand-and-contract pattern; never destructive in the same release).
3. `collect.yml` — cron `0 3,15 * * *` (2×/day) full collection→analysis→report pipeline.
4. `dispatch.yml` — cron hourly: notification dispatcher + retries + critical evaluation.

Configuration

Twelve-factor: all provider keys via env. **Absent key = provider not configured = graceful degradation** (FR-901) — deploys never fail because a paid key is missing.

Rollback

Vercel instant rollback for the app; DB migrations reversible or gated; collection plane is stateless per run — a bad run is re-run after fix, signals are idempotent by content-hash.

24 — Testing Strategy

Status	Approved
Owner	Engineering
Depends on	05 (FR IDs), 10, 12

Tests reference FR/NFR IDs. CI green is the merge gate (23).

Layers

1. **Unit** — diff extraction, price-token parsing, hash/dedupe, tz/DST conversion (property-based tests around DST transitions — FR-702 is where scheduling bugs live), significance rubric mapping, quota budgeter.
2. **Collector contract tests** — each `SignalProvider` runs against **recorded fixtures** (saved HTML/JSON responses, refreshed deliberately, never live-fetched in CI): asserts normalized `Signal` shape, hash stability, robots/allow-list refusal paths, blocked-state handling. Fixtures include Greenhouse/Lever/Ashby board payloads, RSS variants, Wayback CDX responses, GitHub API pages.
3. **Compliance tests (fail-closed)** — fetch layer refuses: non-allow-listed domain, robots-disallowed path, private-IP/SSRF targets, missing UA. These are security tests (21) and block merge.
4. **AI-layer tests** — - *Schema/validator tests*: every task's output validated; `verified-without-sources` rejected; `insufficient_evidence` path covered. - *Golden-set evals*: ~30 curated signal fixtures with expected significance level and label; run on every prompt-template change; regression threshold enforced. (Prompts are versioned code — they get tests like code.) - *Injection suite*: scraped-content fixtures containing adversarial instructions must not alter output structure (21§2).
5. **API contract tests** — responses validated against OpenAPI; label invariants (20 \$invariants) asserted on every claim-bearing endpoint.
6. **E2E (Playwright)** — the release-criteria journeys (02§4): onboard→confirm competitors; view baseline; configure notifications (HH:MM + quiet hours); receive test digest; evidence-click ≤ 1 interaction.
7. **Scheduled-pipeline smoke** — a nightly Actions run against 2 fixture companies end-to-end, writing to a test DB; failure opens an issue.

Coverage stance

No vanity % target. Required: every FR has ≥ 1 mapped test; compliance suite exhaustive; AI golden set current with prompts.

25 — Roadmap

Status	Approved
Owner	Product
Depends on	02, 29

V1 — MVP (6-week build)

Week	Milestone (exit criterion)
1	Skeleton + compliance fetch layer + schema (fetch refusal paths tested)
2	PageDiff collector + Wayback backfill (queryable historical changes for 5 real companies)
3	Hiring (ATS) + Content collectors (hiring-over-time series per competitor)
4	Analysis engine on free-LLM cascade (3–7 labeled insights from a week of signals)
5	Understanding + Discovery + onboarding UX (URL→confirmed set < 10 min)
6	Baseline report, CEO Brief + digests, Email+Telegram at HH:MM, Decision Workspace (actions, Inbox, Roadmap-lite, log), Market Timeline, demo + coverage post-mortem

V1.x — hardening (post-demo)

GitHub + News + TechEcosystem collectors fully wired to preferences; monthly/quarterly reports; export polish; feedback-threshold tuning live.

The V1 discipline (binding)

A finished, polished V1 beats an unfinished platform with 100 planned features. V1 completeness is defined by 02§4 release criteria — onboarding, competitor discovery, the intelligence workspace, the CEO Brief, and the Decision Workspace, all polished. Nothing enters V1 scope after Week 2 without removing something of equal size (ADR-012).

V2 — after validated interest

Paid provider slots activated (Crunchbase funding, firmographics, licensed reviews, tech-stack API); **GitHub Issues export from the Decision Workspace**; **Personal Intelligence ranking** (digest/Brief reordering learned from decision + feedback stream — 13§5); Slack + Discord + WhatsApp channels; team seats + shared workspaces (P2 monetization per Vision review); pgvector search; custom cron-like schedules; report white-labeling.

V3 — analyst maturity

Feedback-trained significance models per segment; per-founder attention models (full Personal Intelligence); Jira/Linear export; multi-language monitoring; "what we said vs. what happened" auto-retrospectives expanded; customer-facing API; mobile push; CRM/Slack workflow integrations.

Explicit never (Manifesto)

Autonomous actions on the user's behalf; authenticated/paywalled collection; reselling collected data.

26 — Coding Standards

Status	Approved
Owner	Engineering

Stack & style

- TypeScript strict everywhere (app + jobs). Python permitted only in isolated analysis scripts if justified by an ADR.
- ESLint + Prettier, enforced in CI; no warnings on main.
- Naming: `SignalProvider` implementations `<Source><Category>Provider`; LLM task templates `tasks/<task_name>@v<N>.md`.

Architecture rules (lint-enforced where possible)

- Interfaces at the boundaries:** collectors, interpretation, delivery channels — implementations never imported directly outside their registry (ADR-003).
- No LLM/network calls in the interactive plane** except `Search` (22§1).
- No hard-coded colors/spacing** in components — tokens only (18).
- Evidence labels are types**, not strings scattered: `type EvidenceLabel = "verified" | "inference" | "recommendation"` — single source in `core/types`.
- Untrusted scraped text is typed `UntrustedText` and cannot enter a prompt template except through the delimiting wrapper (21§2).

Process

- Trunk-based; PRs small; every behavior change updates the matching doc **in the same PR** (NFR-7.3) and references FR/ADR IDs in the description.
- Migrations: expand-and-contract; reviewed like code.
- Commits: Conventional Commits.
- Prompts, fixtures, and golden sets live in-repo and are reviewed like code (24§4).

Documentation of record

`/docs` (this set) is authoritative. Code comments explain *why*, not *what*. Any decision that would surprise a future maintainer becomes an ADR in 28.

27 — Data Source Catalog

Status	Approved — authoritative allow-list (21§1). Adding a source = PR to this file with legal-basis row.
Owner	Engineering
Depends on	12, 21

Legend — Refresh: per collection run (2×/day) unless noted. Legal basis reviewed at add time; free-tier limits verified at build time (they change).

Category: Website / product / pricing (grade A)

Source	Provides	Cost	Refresh	Legal considerations
Competitor public pages (home, pricing, product, changelog)	Content diffs, price tokens	Free	2×/day	robots.txt enforced; public non-authenticated only; polite rate limits; identifying UA
sitemap.xml	Watchable-page discovery	Free	Weekly	Published for crawling by definition; robots still checked
Wayback Machine CDX API	Historical snapshots (backfill)	Free	Once per competitor-add	Public archive API; attribution in evidence view; ≤ 12 snapshots/page cap

Category: Hiring (grade A)

Source	Provides	Cost	Refresh	Legal considerations
Greenhouse public board API (boards.greenhouse.io/v1/boards/{slug}/jobs)	Structured postings	Free	2×/day	Public, unauthenticated, documented for public consumption
Lever postings API (api.lever.co/v0/postings/{slug})	Structured postings	Free	2×/day	Same
Ashby public job board JSON	Structured postings	Free	2×/day	Same
Company careers pages	Fallback diffs	Free	2×/day	robots.txt enforced
✕ Job-board aggregators (LinkedIn/Indeed etc.)	—	—	—	Excluded: ToS prohibits scraping

Category: Content (grade A)

Source	Provides	Cost	Refresh	Legal considerations
Blog/changelog RSS/Atom	Posts, releases, cadence	Free	2×/day	Feeds are published for syndication; excerpts stored, full text linked

Category: Engineering (grade B)

Source	Provides	Cost	Refresh	Legal considerations
GitHub REST API (token)	Releases, repos, stars/forks, public dependency files	Free (5k req/hr)	2×/day	API ToS-compliant; public repos only

Category: Business / funding (grade B — labeled *partial coverage*)

Source	Provides	Cost	Refresh	Legal considerations
Google News RSS	Press mentions, announcements	Free	2×/day	Headlines/links + snippets; full articles linked, not stored (copyright)
Competitor press/newsroom pages	Official announcements	Free	2×/day	robots enforced
SEC EDGAR full-text search	US filings (Form D, S-1...)	Free	Daily	Public government data; fair-access rate limits respected
🔒 Crunchbase API	Structured funding	Paid slot (V2)	—	Licensed; FundingProvider(paid) stub in V1

Category: Technology ecosystem (grade A — global shared targets)

Source	Provides	Cost	Refresh
Official blogs/changelogs/status feeds: OpenAI, Google (Gemini), Anthropic (Claude), Groq, Cerebras, OpenRouter, DeepSeek, Mistral, Meta (Llama), xAI, Perplexity	Models, pricing, context windows, SDK/API/MCP changes, deprecations	Free	2×/day
Legal: official first-party publications; robots enforced; collected once, shared across all users.			

Category: Customer sentiment (grade C — labeled *narrow slice*)

Source	Provides	Cost	Refresh	Legal considerations
Hacker News (Algolia API)	Mentions, discussion sentiment	Free	2×/day	Fully open public API
Reddit API (free tier)	Subreddit mentions	Free (rate-limited)	Daily	Within free-tier non-commercial terms — re-verify terms at build time and before any commercial launch
✕ G2 / Capterra	—	—	—	Excluded: ToS prohibits scraping; ReviewSignalProvider paid/licensed slot only

Category: Firmographics (grade D)

Source	Status
🔒 Apollo/Clearbit/Lead4words APIs - Product pages	Excluded from V1 — no compliant free source; visible locked slot; outputs would be labeled third-party-sourced

Search assist (discovery only)

Source	Provides	Cost	Legal
Google Programmable Search JSON API	Discovery queries	Free — 100 queries/day	Official API; quota budgeting per ADR-005
✕ Scraping search engines directly	—	—	Excluded (ToS)

28 — Product Decisions (ADR Log)

Status	Living document — append-only
Owner	Product + Architecture

Format: Context → Decision → Consequences. Future changes must be consistent with these or supersede them explicitly.

ADR-001 — Wayback backfill is a day-one requirement, not a nice-to-have

Context: Cold start: a change-detection product has nothing to say in week one; the Vision review identified longitudinal history as the product's only compounding asset. **Decision:** Backfill up to 24 months of pricing/home snapshots via the Wayback CDX API at competitor-add; the Baseline Report leads with longitudinal facts. **Consequences:** Day-one value; the moat asset starts accruing immediately; `backfilled` flag distinguishes archive-derived evidence in the UI.

ADR-002 — Recommendations are grounded in public knowledge of both sides only

Context: Truly personalized advice ("discount your at-risk accounts") requires private context the system doesn't have; faking it is the fastest trust-killer. Zero-effort onboarding forbids questionnaires. **Decision:** The user's own site is ingested through the identical collection pipeline; all recommendations condition on public profiles of both sides. Deeper context is earned later via optional, per-insight micro-questions (V2), never demanded upfront. **Consequences:** Honest, defensible recommendations; Moment-4 scope is narrower but real; symmetric pipeline reuse simplifies architecture.

ADR-003 — Provider interfaces per signal category; configuration-driven registration

Context: MVP constraint: paid APIs must be addable later without refactoring; graceful degradation must be real, not an untested branch. **Decision:** One interface per category (and for LLMs and delivery channels); a provider activates iff its env key exists; absent → free provider → explicit coverage gap. Degradation is thus the daily default path. **Consequences:** Zero-code paid upgrades; honest locked slots in UI; slight up-front abstraction cost accepted.

ADR-004 — Evidence buckets in UI; no numeric confidence

Context: LLM numeric confidence is poorly calibrated; "82% confident" is false precision that ships the exact trust failure the product exists to avoid. **Decision:** User-facing evidence is three labels (verified / inference / recommendation) plus source counts ("corroborated by 3 sources"). Numeric scores remain internal (discovery ranking, gating). **Consequences:** Defensible trust surface; simpler UI; internal scores can mature into calibrated ones before ever being exposed.

ADR-005 — Quota is an architectural input, not an ops concern

Context: Free tiers impose RPM/RPD ceilings (LLM) and 100 searches/day (discovery). **Decision:** Central budgeters for LLM tokens/requests and search queries; batching + content-hash caching mandatory; overflow defers visibly ("queued — quota resets 00:00 UTC"), never drops or degrades silently. **Consequences:** Predictable behavior at \$0; the deferral state is designed UI, not an error.

ADR-006 — UTC storage, IANA per-user conversion, HH:MM wall-clock semantics

Context: Users pick any HH:MM; DST must not shift their 08:00 digest. **Decision:** Store UTC everywhere; `users.tz` (IANA) converts at dispatch time; hourly dispatcher evaluates wall-clock windows; property-based DST tests required (24\$1). **Consequences:** Correct scheduling year-round; ±5 min SLA achievable on an hourly tick with minute-level enqueue.

ADR-007 — G2/Capterra and job aggregators excluded; compliance is fail-closed

Context: Their ToS prohibit scraping; the Manifesto makes compliance a floor. **Decision:** Excluded entirely; visible licensed-provider stubs instead; allow-list (27) is the only fetchable surface and is fail-closed in code. **Consequences:** Narrower sentiment coverage, honestly labeled — coverage honesty is a feature (17), and the exclusion itself demonstrates the product's values.

ADR-008 — GitHub Actions as the scheduler

Context: Free-tier serverless can't run long jobs; the MVP needs a free, reliable, auditable scheduler. **Decision:** Public-repo Actions cron for collection (2×/day) and dispatch (hourly). Runs double as a public collection audit log. **Consequences:** \$0, transparent, portfolio-visible; cron jitter accepted within the days/weeks product timescale (NFR-8); migration path to a queue service in V2 if validated.

ADR-009 — Two-plane architecture (interactive vs. collection)

Context: Function execution ceilings; precompute-over-compute performance stance (22). **Decision:** The web app never performs collection/interpretation (Search excepted, streamed); all heavy work runs scheduled and writes stored artifacts. **Consequences:** Fast pages at \$0; clean failure isolation; Search is the one latency-sensitive LLM path and is budgeted separately.

ADR-010 — Postgres FTS before vectors

Context: Signals/insights are already structured; MVP corpus is small. **Decision:** SQL + tsvector retrieval in V1; pgvector deferred to V2 (25). **Consequences:** Simpler stack, zero added infra; revisit when corpus or query complexity outgrows FTS.

ADR-011 — Evidence-derived conviction scores on recommendations (amends ADR-004)

Context: Reviewer feedback: score recommendations, not just competitors ("Confidence 93% — because hiring trend, GitHub activity, documentation updates support it"). ADR-004 bans numeric confidence because LLM *self-reported* confidence is uncalibratable. **Decision:** Recommendations carry a **conviction score computed deterministically from evidence** — the count and agreement of independent supporting signal families — always displayed with those factors enumerated. This is explainable arithmetic over observations, not model self-assessment, so it does not violate ADR-004's rationale; ADR-004 continues to ban *verbalized LLM confidence* everywhere. UI defaults to buckets (High/Medium/Low +

factors); the numeric form may be exposed once the weekly audit (10\$5) shows the score tracks outcome quality. **Consequences:** Recommendations become rankable (Decisions Inbox, CEO Brief prioritization) and *more* trustworthy, because the number always arrives with its reasons.

ADR-012 — Decision Workspace in V1, deliberately lite

Context: Reviewer identified the Decision Workspace as the biggest missing feature — without it the loop ends at reading. Same reviewer's strongest advice: finish V1; don't expand endlessly. **Decision:** V1 ships the six decision actions, Inbox, Opportunities, three-stage Roadmap-lite, and the Decision log — and nothing more. GitHub Issues export, assignees, due dates, comments, and any PM-tool surface are V2+. GitHub export renders as a visible disabled slot (consistent with ADR-003's honest-locked-slot pattern). Zero-sum scope rule: nothing enters V1 after Week 2 without an equal removal. **Consequences:** The loop closes (read → decide) at small engineering cost (one table, one board, one bar component); the North Star becomes directly measurable; scope pressure has a named gate.

ADR-013 — Command-center UI; streamed-reasoning onboarding

Context: Reviewer: "Bloomberg Terminal meets Linear — one large command center, not cards-cards-cards"; onboarding should show the AI working, like Perplexity/ChatGPT reasoning. **Decision:** Three-pane keyboard-driven workspace with ⌘K palette as primary navigation; card-grid defaults banned (17). Onboarding streams genuine pipeline steps with real artifacts per step — never fake progress animation. **Consequences:** Higher UI engineering bar accepted for V1 polish (per ADR-012's finish-V1 discipline); the onboarding stream doubles as the product's first trust demonstration.

Status	Approved — honesty is a feature; this document is user-facing in spirit (mirrored in-app under Settings → Data & sources)
Owner	Product

The MVP deliberately does **not** do the following. Each limitation is a decision (28), not an oversight.

Data & coverage

- 1. **Public signals only.** No authenticated, paywalled, or CAPTCHA-protected content — ever. Anything a logged-out browser can't see, we can't either.
- 2. **No paid data APIs.** Crunchbase, BuiltWith-class, Apollo/Clearbit-class, licensed review data: visible locked provider slots, not silent dependencies (FR-901).
- 3. **Funding/business coverage is partial** — free news + EDGAR miss unannounced or non-US events; labeled *partial* in the UI.
- 4. **Customer sentiment is a narrow slice** — HN + limited Reddit only; **G2/Capterra excluded by their ToS** (ADR-007). Sentiment insights are labeled *narrow coverage* and are inferences, not verified metrics.
- 5. **No verified firmographics** (employee counts, revenue estimates). The slot exists; the data partner doesn't yet.
- 6. **Tech-stack detection is passive inference** from public page source only — labeled single-source inference.
- 7. **English-language, primarily US/global-SaaS competitor sets.** Multi-language monitoring is V3.
- 8. **Coverage gaps are visible, not hidden.** A blocked or dead source shows as a gap state; "no news" and "couldn't look" are never conflated.

Intelligence & trust

- 9. **Every AI conclusion is labeled.** Verified / AI Inference / Reasoned Recommendation — no exceptions. The model's *self-reported* confidence is never shown (ADR-004); recommendations instead carry an evidence-derived conviction with its supporting signal families enumerated (ADR-011), shown as High/Medium/Low buckets until calibration is demonstrated.
- 10. **Recommendations use public context of both sides only** (ADR-002). The system does not know private revenue, pipeline, or strategy, and never pretends to.
- 11. **The system may say "insufficient evidence."** It will not fill gaps with plausible text.
- 12. **Significance thresholds start as heuristics** (13 rubric) tuned by simple feedback — not a trained model (V3).

Product scope

- 13. **Timescale is hours-to-days.** Not real-time; no trading/market-sentiment use.
- 14. **Advisory only.** No autonomous actions on the user's behalf — a permanent non-goal, not an MVP gap.
- 15. **Channels: Email + Telegram only.** WhatsApp/Slack/Discord/push are future (15§1).
- 16. **Single-user workspaces.** No teams, roles, or sharing permissions (V2).
- 17. **Custom cron-like schedules not yet supported** — daily/weekdays/weekly/monthly/instant-critical only (15§3). 17a. **Decision Workspace is deliberately lite** — no GitHub/Jira/Linear export yet (visible disabled slot), no assignees/due dates/comments; it bridges to your PM tools, it doesn't replace them (ADR-012). 17b. **Personal Intelligence is not yet active** — V1 records your decisions and feedback; personalized ranking of Briefs/digests begins in V2 (13§5).
- 18. **Search answers only from collected evidence** plus one-off quota-bounded analyses; it is not a general web assistant.

Infrastructure honesty

- 19. **Free-tier quotas shape behavior:** interpretation can defer to the next run under quota pressure ("queued — resets 00:00 UTC"); discovery search is capped at 100 queries/day shared (ADR-005). Deferral is visible, never silent loss.
- 20. **Storage discipline:** raw HTML kept only for latest snapshots; history is stored as extracted text + structured diffs (NFR-5.1).
- 21. **Scheduling tolerance ± 5 min** on an hourly dispatcher (ADR-006/008) — appropriate to the product's timescale.
- 22. **Built for demonstration and learning first.** Scale hardening (queues, caching layers, SLOs) begins only after user interest is validated — per the MVP constraints in 01§2.

Status	Approved (added in Rev. 1.1 — reviewer feedback)
Owner	Product, Engineering
Depends on	13, 14, 17; ADR-012

Why this exists

Intelligence that ends at a report is still information. The Manifesto's test — *does this help the founder decide what to build, what to ignore, and what to do next?* — is only fully passed when the product captures the decision itself. The loop therefore extends:

```
Website → Competitors → Signals → AI Analysis → Recommendations
      → Decision Workspace → Roadmap → Execution
```

The product stops being something founders *read* and becomes something that *changes what they do*.

1. Decision actions (V1 — on every recommendation and insight)

Action	Effect
Save as Opportunity	Creates a Decision record (opportunity) linked to the insight + its full evidence chain
Ignore	Records ignored + optional reason; feeds the significance/personalization loop (13§5) — the strongest negative-preference signal available
Create Feature	Creates a feature Decision with an AI-drafted one-paragraph brief (problem, evidence, competitor context) — editable
Add to Roadmap	Places a Decision into the lightweight Roadmap board (below)
Review Later	Snoozes to a chosen date; resurfaces in that day's CEO Brief
Export to GitHub Issues	Future (V2) — visible disabled slot, consistent with the locked-provider pattern (ADR-003)

Buttons render inline on insight cards and inside reports (<DecisionBar>, 19). One click, no modal for the simple actions.

2. The Workspace views

- **Inbox** — undecided recommendations, ranked by significance × conviction.
- **Opportunities** — saved items with evidence chains; sortable by conviction, recency, competitor.
- **Roadmap (lite)** — three columns: *Considering* → *Planned* → *Building*. Deliberately not a project-management tool (Manifesto non-goal: not a CRM/PM suite); it is the bridge from intelligence to the founder's real tools. Each card retains its provenance: decision → insight → signals → sources (FR-802 extended).
- **Decision log** — append-only history: what was decided, when, on what evidence. Powers the "what we said vs. what happened" retrospectives (14) — now extended to "what *you* decided vs. what happened."

3. Closing the loop

Decisions are the product's highest-value feedback: `saved/created/roadmapped` ⇒ strong positive weight; `ignored` ⇒ strong negative weight — both feed significance thresholds (13§5) and Personal Intelligence (V2/V3, 25). The post-validation North Star (Weekly **Actioned** Insights, 01§7) is measured directly from this table instead of proxies.

4. Schema (adds to 09)

```
create table decisions (
  id uuid primary key default gen_random_uuid(),
  company_id uuid references companies not null,
  insight_id uuid references insights not null,
  action text not null check (action in
    ('opportunity','ignored','feature','roadmap','review_later')),
  stage text check (stage in ('considering','planned','building')), -- roadmap items
  brief text, -- AI-drafted, user-edited (Create Feature)
  review_at date, -- Review Later
  reason text, -- optional, on Ignore
  created_at timestamptz default now()
);
```

5. API (adds to 20)

```
POST /insights/:id/decisions { action, stage?, review_at?, reason? }
GET  /companies/:id/decisions ?action=&stage=&cursor=
PATCH /decisions/:id { stage | brief | review_at }
GET  /companies/:id/decisions/log
```

6. V1 scope guard (reviewer's advice: finish V1, don't expand endlessly)

V1 ships: the six actions (GitHub export as a disabled slot), Inbox, Opportunities, Roadmap-lite, Decision log. V1 does **not** ship: GitHub/Jira/Linear export, assignees, due dates, comments, or anything that turns this into a PM tool. See ADR-012.